

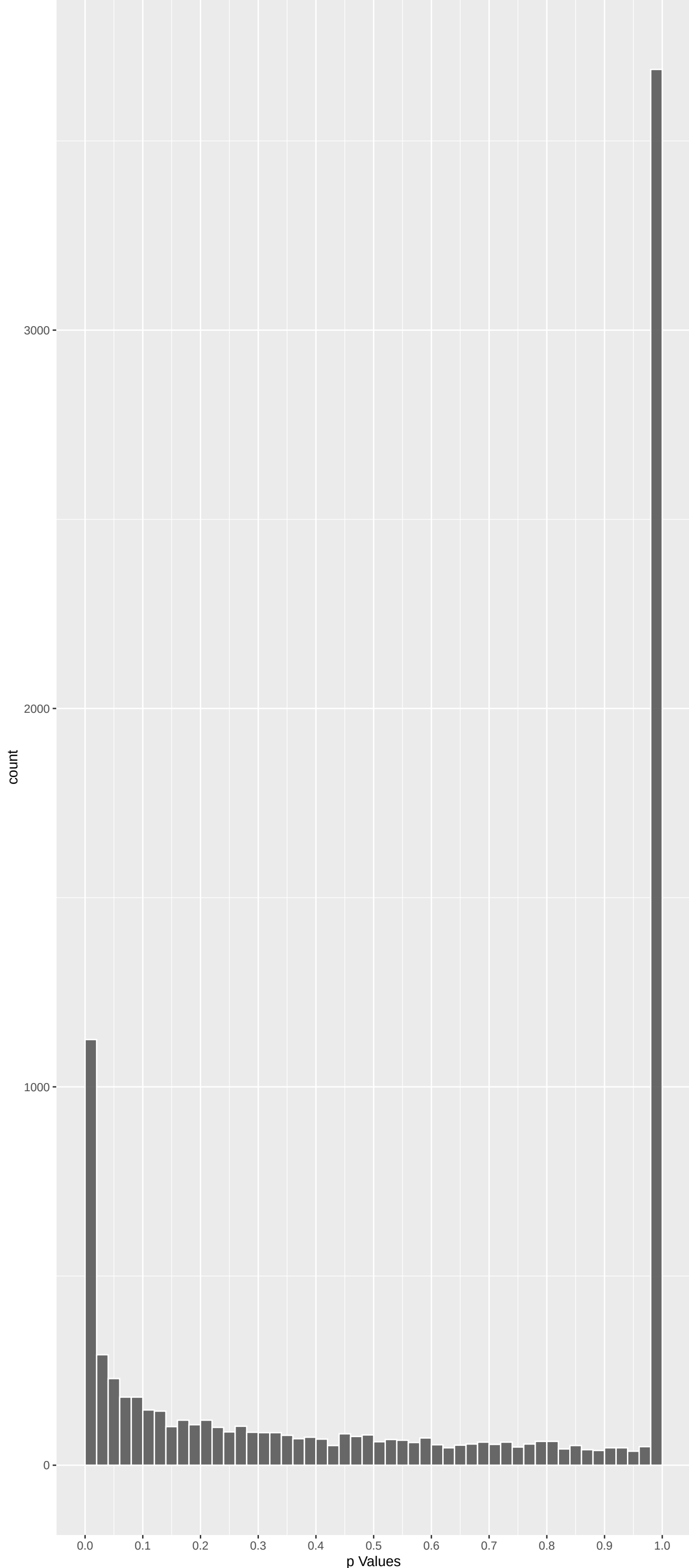
Top genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
TRPM5	6.972544	9.337787e-12	1.506e-07	2.093e-03
EP400	6.653860	8.565107e-11	4.777e-07	2.093e-03
RIMBP2	6.648480	8.884070e-11	4.777e-07	2.093e-03
TGFBRAP1	6.553312	1.688241e-10	6.488e-07	2.093e-03
CAMSAP1	6.527138	2.011150e-10	6.488e-07	2.093e-03
AKAP9	-6.386429	5.094127e-10	1.369e-06	3.682e-03
LRRC53	-6.255679	1.187370e-09	2.736e-06	6.123e-03
ZSCAN20	-6.181392	1.906161e-09	3.416e-06	6.123e-03
NSD2	6.196917	1.727397e-09	3.416e-06	6.123e-03
TRANK1	6.114494	2.905912e-09	4.687e-06	7.561e-03
MUC20	-6.068554	3.872005e-09	5.678e-06	8.326e-03
CLCNKA	-6.004749	5.748830e-09	7.727e-06	1.039e-02
MAP3K15	5.949380	8.074781e-09	9.762e-06	1.125e-02
ADH6	-5.941502	8.472688e-09	9.762e-06	1.125e-02
KCNK18	5.915970	9.897771e-09	1.064e-05	1.126e-02
PER2	5.885225	1.192538e-08	1.186e-05	1.126e-02
MROH2A	5.877413	1.250186e-08	1.186e-05	1.126e-02
ADGRB1	5.851496	1.461517e-08	1.310e-05	1.127e-02
IRAG2	-5.840286	1.563338e-08	1.327e-05	1.127e-02
SHANK2	5.811236	1.860398e-08	1.500e-05	1.210e-02
NEK5	-5.774731	2.312280e-08	1.776e-05	1.364e-02
TET1	-5.684242	3.941822e-08	2.890e-05	2.119e-02
LRPPRC	-5.619728	5.737737e-08	3.856e-05	2.592e-02
FANCM	-5.624284	5.588360e-08	3.856e-05	2.592e-02
ATAD5	-5.574405	7.451303e-08	4.680e-05	2.796e-02

Top genes by Q-Value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
EP400	6.653860	8.565107e-11	4.777e-07	2.093e-03
TGFBRAP1	6.553312	1.688241e-10	6.488e-07	2.093e-03
CAMSAP1	6.527138	2.011150e-10	6.488e-07	2.093e-03
TRPM5	6.972544	9.337787e-12	1.506e-07	2.093e-03
RIMBP2	6.648480	8.884070e-11	4.777e-07	2.093e-03
AKAP9	-6.386429	5.094127e-10	1.369e-06	3.682e-03
LRRC53	-6.255679	1.187370e-09	2.736e-06	6.123e-03
ZSCAN20	-6.181392	1.906161e-09	3.416e-06	6.123e-03
NSD2	6.196917	1.727397e-09	3.416e-06	6.123e-03
TRANK1	6.114494	2.905912e-09	4.687e-06	7.561e-03
MUC20	-6.068554	3.872005e-09	5.678e-06	8.326e-03
CLCNKA	-6.004749	5.748830e-09	7.727e-06	1.039e-02
MAP3K15	5.949380	8.074781e-09	9.762e-06	1.125e-02
ADH6	-5.941502	8.472688e-09	9.762e-06	1.125e-02
PER2	5.885225	1.192538e-08	1.186e-05	1.126e-02
KCNK18	5.915970	9.897771e-09	1.064e-05	1.126e-02
MROH2A	5.877413	1.250186e-08	1.186e-05	1.126e-02
ADGRB1	5.851496	1.461517e-08	1.310e-05	1.127e-02
IRAG2	-5.840286	1.563338e-08	1.327e-05	1.127e-02
SHANK2	5.811236	1.860398e-08	1.500e-05	1.210e-02
NEK5	-5.774731	2.312280e-08	1.776e-05	1.364e-02
TET1	-5.684242	3.941822e-08	2.890e-05	2.119e-02
LRPPRC	-5.619728	5.737737e-08	3.856e-05	2.592e-02
FANCM	-5.624284	5.588360e-08	3.856e-05	2.592e-02
ATAD5	-5.574405	7.451303e-08	4.680e-05	2.796e-02

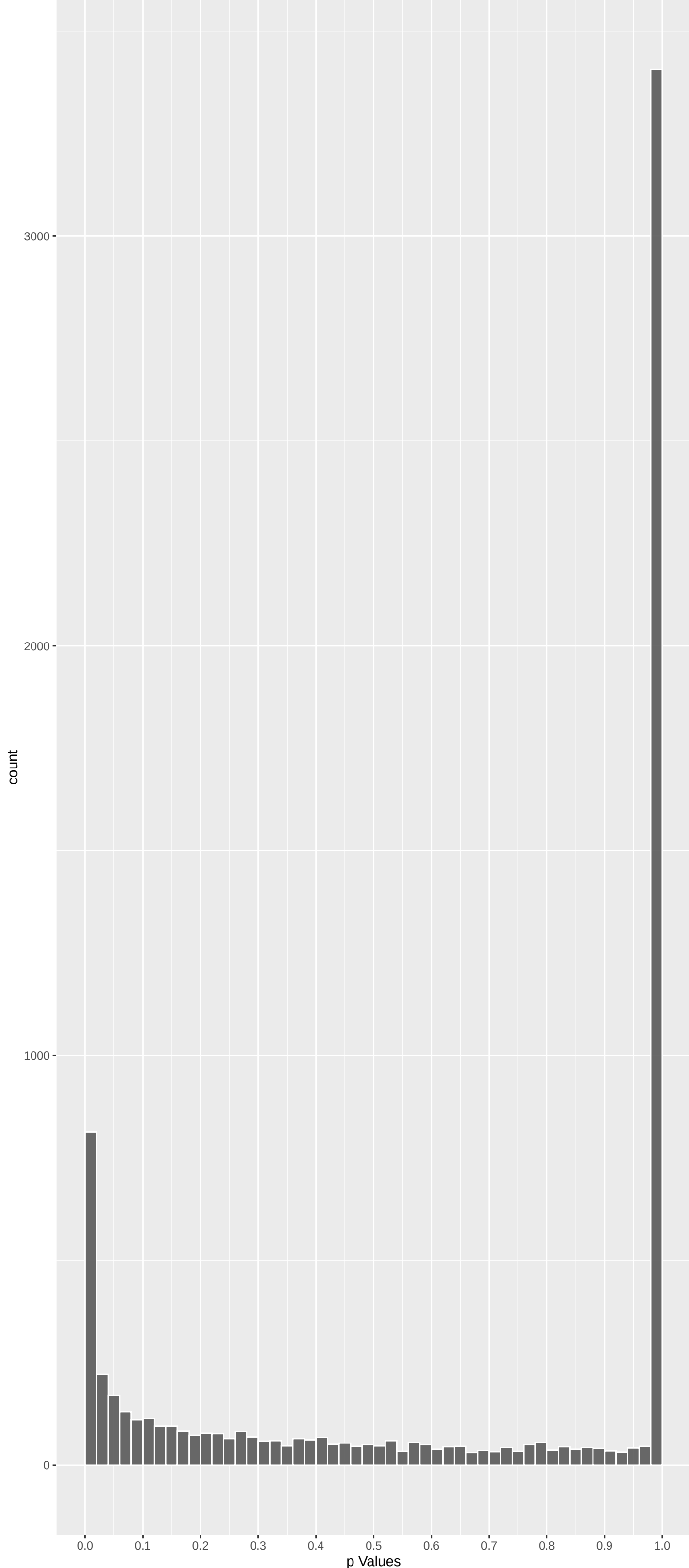
Positive Rho Non-permuted



Top Positive genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
TRPM5	6.972544	9.337787e-12	1.506e-07	2.093e-03
EP400	6.653860	8.565107e-11	4.777e-07	2.093e-03
RIMBP2	6.648480	8.884070e-11	4.777e-07	2.093e-03
TGFBRAP1	6.553312	1.688241e-10	6.488e-07	2.093e-03
CAMSAP1	6.527138	2.011150e-10	6.488e-07	2.093e-03
NSD2	6.196917	1.727397e-09	3.416e-06	6.123e-03
TRANK1	6.114494	2.905912e-09	4.687e-06	7.561e-03
MAP3K15	5.949380	8.074781e-09	9.762e-06	1.125e-02
KCNK18	5.915970	9.897771e-09	1.064e-05	1.126e-02
PER2	5.885225	1.192538e-08	1.186e-05	1.126e-02

Negative Rho Non-permuted



Top Negative genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
AKAP9	-6.386429	5.094127e-10	1.369e-06	3.682e-03
LRRC53	-6.255679	1.187370e-09	2.736e-06	6.123e-03
ZSCAN20	-6.181392	1.906161e-09	3.416e-06	6.123e-03
MUC20	-6.068554	3.872005e-09	5.678e-06	8.326e-03
CLCNKA	-6.004749	5.748830e-09	7.727e-06	1.039e-02
ADH6	-5.941502	8.472688e-09	9.762e-06	1.125e-02
IRAG2	-5.840286	1.563338e-08	1.327e-05	1.127e-02
NEK5	-5.774731	2.312280e-08	1.776e-05	1.364e-02
TET1	-5.684242	3.941822e-08	2.890e-05	2.119e-02
LRPPRC	-5.619728	5.737737e-08	3.856e-05	2.592e-02

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NADH Dehydrogenase Complex Assembly (GO:0140005)	-0.29618573	48	1.297e-12	3.494e-09	TMEM126A:53 NDUFB8:57 NDUFS3:70 NDUFAF1:162 TMEM126B:191 NDUFA11:226
Mitochondrial Respiratory Chain Complex	-0.29618573	48	1.297e-12	3.494e-09	TMEM126A:53 NDUFB8:57 NDUFS3:70 NDUFAF1:162 TMEM126B:191 NDUFA11:226
Mitochondrial Cation Transmembrane Transport	0.12266293	273	3.811e-12	6.844e-09	TRPM5:1 KCNK18:9 KCNF1:67 TRPV6:104 KCNV2:145 KCNGA:148
Chemical Synaptic Transmission (GO:00072)	0.12281779	255	1.761e-11	2.372e-08	RIMBP2:3 GABRB3:72 DLGAP2:75 NPTX1:94 APBA2:142 CACNA1B:175
Mitochondrial Gene Expression (GO:0140005)	-0.19166093	100	3.775e-11	4.068e-08	POLRMT:41 PTCO3:205 TFB2M:418 FASTKD1:464 MRPL3:470 FASTKD5:557
Potassium Ion Transmembrane Transport (GO:0032543)	0.16501107	132	6.465e-11	5.806e-08	KCNK18:9 KCNF1:67 LETM1:138 KCNV2:145 KCNGA:148 TMEM175:160
Mitochondrial Inorganic Cation Transmembrane Transport	-0.18954493	96	1.469e-10	1.131e-07	PTCD3:205 MRPL37:470 TSFM:549 MRPL42:559 MRPS30:694 GFM2:733
Inorganic Cation Transmembrane Transport	0.11053430	277	2.937e-10	1.978e-07	TRPM5:1 KCNK18:9 KCNF1:67 TRPV6:104 KCNV2:145 KCNGA:148
tRNA Methylation (GO:0030488)	-0.29655426	36	7.522e-10	4.503e-07	FTSJ1:97 THADA:116 TRMT10A:133 THUMPD2:317 TRMT10B:336 TRMO:516
Potassium Ion Transport (GO:0006813)	0.15915684	119	2.155e-09	1.161e-06	KCNK18:9 KCNF1:67 KCNV2:145 KCNGA:148 TMEM175:160 KCNA10:185
Mitochondrial Respiratory Chain Complex	-0.19687515	77	2.524e-09	1.236e-06	TMEM126A:53 NDUFB8:57 NDUFS3:70 NDUFAF1:162 TMEM126B:191 NDUFA11:226
tRNA Modification (GO:0006400)	-0.20929534	66	4.243e-09	1.905e-06	FTSJ1:97 THADA:116 TRMT10A:133 TRMT10B:336 THUMPD2:317 PUS3:326
DNA Repair (GO:0006281)	-0.09911106	274	1.901e-08	7.879e-06	FANCM:10 BRCA1:34 MBD4:44 BRCA2:49 POLQ:52 RNF168:63
Oxidative Phosphorylation (GO:0006119)	-0.21252989	55	5.099e-08	1.832e-05	NDUFB8:57 NDUFS3:70 NDUFAF1:162 NDUFS5:400 NDUFA12:451 NDUFB5:594
Proton Motive Force-Driven Mitochondrial	-0.23010659	47	4.916e-08	1.832e-05	NDUFB8:57 NDUFS3:70 NDUFAF1:162 NDUFS5:400 NDUFA12:451 NDUFB5:594
Metal Ion Transport (GO:0030001)	0.12571401	157	5.884e-08	1.981e-05	KCNK18:9 KCNF1:67 TRPV6:104 SLC17A4:121 ITPR1:128 KCNA10:185
Regulation Of Heart Rate By Cardiac Cond	0.24912102	39	7.418e-08	2.351e-05	BIN1:376 DSG2:402 CACNB2:477 KCNJ3:535 KCNH2:731 CAV1:870
Proton Motive Force-Driven ATP Synthesis	-0.21570503	51	1.008e-07	3.018e-05	NDUFB8:57 NDUFS3:70 NDUFAF1:162 NDUFS5:400 NDUFA12:451 NDUFB5:594
RNA Methylation (GO:0001510)	-0.20084124	58	1.249e-07	3.366e-05	FTSJ1:97 THADA:116 TRMT10A:133 THUMPD2:317 TFB2M:418 METTL4:430
Aerobic Respiration (GO:0009060)	-0.20831377	54	1.215e-07	3.366e-05	NDUFB8:57 NDUFS3:70 NDUFAF1:162 MTFR2:328 NDUFS5:400 NDUFA12:451
Cardiac Conduction (GO:0061337)	0.22960881	44	1.390e-07	3.566e-05	BIN1:376 DSG2:402 CACNB2:477 KCNJ3:535 KCNH2:731 CAV1:870
Mitochondrial Electron Transport, NADH T	-0.26434647	31	3.542e-07	8.675e-05	NDUFB8:57 NDUFS3:70 NDUFAF1:162 NDUFS5:400 NDUFB5:594 NDUFB10:598
Inorganic Cation Import Across Plasma Membr	0.14688391	96	6.829e-07	1.600e-04	TRPV6:104 CACNA11:172 CACNA1B:175 SCN2A:190 TRPV2:200 KCNJ12:202
Calcium Ion Import Across Plasma Membran	0.24658651	33	9.565e-07	2.147e-04	TRPV6:104 CACNA11:172 CACNA1B:175 SCN2A:190 TRPV2:200 TRPM1:206
Calcium Ion Transport (GO:0006816)	0.13639110	106	1.274e-06	2.747e-04	TRPM5:1 TRPV6:104 ITPR1:128 TRPV2:200 TRPM1:206 ATP2A1:366
Cellular Respiration (GO:0045333)	-0.16406629	71	1.795e-06	3.720e-04	NDUFB8:57 NDUFS3:70 NDUFAF1:162 MTFR2:328 NDUFS5:400 NDUFA12:451
Sister Chromatid Segregation (GO:0000819)	-0.23716087	33	2.335e-06	4.660e-04	SMC4:174 KIFC1:212 ZWINT:245 KIF18A:281 SPAG5:324 NUSAP1:764
Muscle Contraction (GO:0006936)	0.14556917	88	2.433e-06	4.681e-04	TACR2:96 MYH13:114 EMD:132 MYH2:195 KCNJ12:202 LMOD1:384
Double-Strand Break Repair Via Homologou	-0.13968385	106	3.361e-06	6.245e-04	FANCM:10 BRCA1:34 BRCA2:49 RMI1:121 HELQ1:137 CDT3:204
DNA Metabolic Process (GO:0006936)	-0.08079023	275	4.434e-06	7.964e-04	BRCA1:34 MBD4:44 POLQ:52 RMI1:121 ORC1:137 PARP4:179
Anterograde Trans-Synaptic Signaling (GO:009820288)	0.09820288	181	5.532e-06	9.614e-04	GABRB3:72 NPTX1:94 APBA2:142 CACNA1B:175 KCNQ3:193 SLC6A2:334
Double-Strand Break Repair (GO:0006302)	-0.10310421	159	7.642e-06	1.287e-03	BRCA1:34 BRCA2:49 POLQ:52 RNF168:63 RMI1:121 HELQ1:173
Atrial Cardiac Muscle Cell Action Potent	0.38612531	11	9.245e-06	1.509e-03	CACNB2:477 KCNJ3:535 GJC1:741 KCNJ5:1286 KCNN2:1421 KCNQ1:1515
Mitochondrial ATP Synthesis Coupled Elec	-0.17102490	55	1.166e-05	1.847e-03	NDUFB8:57 NDUFS3:70 NDUFAF1:162 NDUFS5:400 NDUFA12:451 NDUFB5:594
Cardiac Muscle Cell Action Potential Inv	0.23839373	28	1.272e-05	1.958e-03	ATPIA1:492 KCNH2:731 GJC1:741 SCN1A:885 NEDD4:1157 CACNA1D:1211
Neuron Projection Morphogenesis (GO:00408)	0.10906808	134	1.361e-05	2.036e-03	ADGRB1:12 CNTN4:123 TRIO:219 DIP2A:231 SHANK3:248 DVL1:278
Monoatomic Ion Transport (GO:0006811)	0.12330009	104	1.440e-05	2.097e-03	SLC17A4:121 PIEZO1:440 CHRNB2:482 SLC15A1:634 TRPM3:651 SLC2A1:678
Visual Perception (GO:0007601)	0.12871250	95	1.496e-05	2.121e-03	CRYGA:42 GPR143:89 TRPM1:206 RD3:226 NXN12:262 RGS9BP:289
Sensory Perception Of Light Stimulus (GO:0007601)	0.12723196	96	1.695e-05	2.341e-03	CRYGA:42 GPR143:89 TRPM1:206 RD3:226 NXN12:262 VAX2:327
Intermediate Filament Organization (GO:0007601)	0.17530162	50	1.825e-05	2.458e-03	BFSFP2:80 KRT80:103 PKP1:168 KRT17:304 KRT12:573 DES:850

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
BENPORATH_ES_WITH_H3K27ME3	0.09261981	982	2.130e-22	1.382e-18	RGS9:16 PLEC:39 FRMPD4:55 FZD10:60 KCNF1:67 NPTX1:94
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	0.11473134	555	3.705e-20	1.202e-16	ADGRB1:12 DLGAP2:75 CADM3:110 CNTN4:123 NETO1:140 APBA2:142
FISCHER_DREAM_TARGETS	-0.08580945	882	9.749e-18	2.108e-14	FANCM:10 ATAD5:12 MIS18BP1:18 DLGAP5:26 FAM161A:29 BRCA1:34
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	0.11819255	420	1.254e-16	2.033e-13	ADGRB1:12 FAM83G:35 FZD10:60 DLGAP2:75 SDK2:85 STUM:157
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	0.13183053	331	2.028e-16	2.632e-13	ADGRB1:12 DLGAP2:75 PRDM16:101 CADM3:110 FAM163B:152 STUM:157
BENPORATH_EED_TARGETS	0.07899000	927	6.377e-16	6.895e-13	PLEC:39 FRMPD4:55 FZD10:60 NPTX1:94 IGFBP3:102 ADRB1:112
REACTOME_NEURONAL_SYSTEM	0.12011302	383	8.765e-16	8.124e-13	KCNK18:9 SHANK2:13 KCNF1:67 GABRB3:72 DLGAP2:75 PPPIA:126
MIKKELSEN_NPC_HCP_WITH_H3K27ME3	0.12858322	331	1.076e-15	8.722e-13	ADGRB1:12 DLGAP2:75 PRDM16:101 CADM3:110 FAM163B:152 C1QL2:159
NIKOLSKY_BREAST_CANCER_7P22_AMP_LICON	0.36234637	38	1.092e-14	7.833e-12	ADAP1:137 TTYH3:255 LFNG:268 INTS1:271 SNR:293 BRAT1:296
MARTENS_TRETINOIN_RESPONSE_UP	0.08756474	676	1.207e-14	7.833e-12	SHANK2:13 CEP170B:41 CHST15:53 FZD10:60 TRIM29:66 KIF1A:68
BENPORATH_SUZ12_TARGETS	0.07483369	918	2.504e-14	1.477e-11	PLEC:39 CYP4A:54 FRMPD4:55 FZD10:60 NPTX1:94 CADM3:110
HAMAI_APOPTOSIS_VIA_TRAIL_UP	-0.08562915	594	1.326e-12	6.619e-10	MIS18BP1:18 DLGAP5:26 AGGF1:31 BRCA1:34 GOLGB1:42 MBD4:44
MARSON_BOUND_BY_E2F4_UNSTIMULATED	-0.08171227	655	1.323e-12	6.618e-10	FANCM:10 DLGAP5:26 KIAA0825:37 BRIP1:47 BRCA2:49 POLQ:52
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	-0.29836499	47	1.495e-12	6.927e-10	NDUFB8:57 NDUFS3:70 NDUFAF1:162 TMEM126B:191 ECSIT:286 NDUFS5:400
BENPORATH_PRC2_TARGETS	0.08465863	580	4.171e-12	1.804e-09	PLEC:39 FRMPD4:55 FZD10:60 NPTX1:94 ADRB1:112 PKP1:168
DODD_NASOPHARYNGEAL_CARCINOMA_DN	-0.05861783	1233	7.641e-12	3.098e-09	TET1:9 FANCM:10 LRPPRC:11 NEDD4:16 DLGAP5:26 KLHL5:33
REACTOME_COMPLEX_I_BIOGENESIS	-0.28143182	48	1.542e-11	5.887e-09	NDUFB8:57 NDUFS3:70 NDUFAF1:162 TMEM126B:191 NDUFA11:226 ECSIT:286
REACTOME_SIGNALING_BY_GPCR	0.07841176	629	2.478e-11	8.930e-09	RGS9:16 SSTR4:21 OPN3:45 FZD10:60 ADORA1:64 GPR143:89
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERAC	0.11963487	261	3.151e-11	1.076e-08	SSTR4:21 ADORA1:64 GABRB3:72 TACR2:96 ADRB1:112 OPR1:203
YOSHIMURA_MAPK8_TARGETS_UP	0.05887599	1109	5.824e-11	1.889e-08	RGS9:16 SSTR4:21 AFDN:23 RNF112:33 CRYGA:42 PTPRE:58
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMP_LICO	0.16394045	128	1.566e-10	4.840e-08	ZBTB46:78 CABLES2:161 OPR1:203 TCF15:215 TPD52L2:333 SYCP2:336
REACTOME_KERATINIZATION	0.19926870	85	2.199e-10	6.486e-08	DSG3:46 KRT80:103 KLK13:127 TGM5:129 PKP1:168 TGM1:173
JOHNSTONE_PARVB_TARGETS_3_DN	-0.06695370	783	2.509e-10	7.079e-08	SFRBP1:17 MIS18BP1:18 NABP:25 DLGAP5:26 BRIP1:47 RTKN2:48
RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE	-0.07617228	596	2.640e-10	7.137e-08	LRPPRC:11 NEDD4:16 DLGAP5:26 AGGF1:31 BRCA1:34 BUB1B:65
REACTOME_MUSCLE_CONTRACTION	0.13350648	187	3.248e-10	8.429e-08	KCNK18:9 NEB:24 MYBPC2:36 ITPR1:128 CACNA11:172 SCN2A:190
KEGG_CALCIIUM_SIGNALING_PATHWAY	0.13887359	171	3.940e-10	9.832e-08	TACR2:96 ADRB1:112 ITPR1:128 ADCY1:135 CACNA11:172 CACNA1B:175
WP_GPCRS_CLASS_A_RHODOPSINLIKE	0.12325232	211	6.672e-10	1.603e-07	SSTR4:21 OPN3:45 ADORA1:64 ADRB1:112 GPR20:113 GPR20:1503
FLORIO_NEOCORTEX_BASAL_RADIAL_GLIA_DN	-0.13264756	179	9.807e-10	2.272e-07	MIS18BP1:18 DLGAP5:26 BRCA1:34 BRIP1:47 BRCA2:49 POLQ:52
GEORGES_TARGETS_OF_MIR192_AND_MIR215	-0.06320558	816	1.129e-09	2.525e-07	FANCM:10 ATAD5:12 MIS18BP1:18 DST:24 NABP2:25 DLGAP5:26
BLANCO_MLO_BRONCHIAL_EPITHELIAL_CELLS_I	-0.13739981	163	1.494e-09	3.231e-07	IRAC2:7 ATAD5:12 FAM161A:29 BRCA1:34 CLSPN:58 BUB1B:65
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	-0.17251676	101	2.144e-09	4.348e-07	LRPPRC:11 NDUFB8:57 NDUFS3:70 NDUFAF1:162 TMEM126B:191 NDUFA11:226
REACTOME_MITOCHONDRIAL_TRANSLATION	-0.18073586	92	2.131e-09	4.348e-07	PTCD3:205 MRPL37:470 TSFM:549 MRPL42:559 MRPS35:643 MRPS30:694
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	-0.19190103	81	2.412e-09	4.743e-07	LRPPRC:11 NDUFB8:57 NDUFS3:70 NDUFAF1:162 TMEM126B:191 NDUFA11:226
MEISSNER_BRAIN_HCP_WITH_H3K4ME3_AND_H3K27M	0.05561849	1024	2.502e-09	4.774e-07	ELFN1:15 FZD10:60 KCNF1:67 PRDM16:101 INAVA:117 COL12A1:141
REACTOME_GPCR_LIGAND_BINDING	0.08631311	398	3.887e-09	7.206e-07	SSTR4:21 OPN3:45 FZD10:60 ADORA1:64 GPR143:89 TACR2:96
REACTOME_FORMATION_OF_THE_CORNFIED_ENVE	0.19167027	78	4.938e-09	8.899e-07	DSG3:46 KRT80:103 KLK13:127 TGM5:129 PKP1:168 TGM1:173
REACTOME_POTASSIUM_CHANNELS	0.17573028	90	8.493e-09	1.489e-06	KCNK18:9 KCNF1:67 KCNV2:145 KCNGA:148 KCNA10:185 KCNQ3:193
REACTOME_VOLTAGE_GATED_POTASSIUM_CHANNEL	0.25155556	42	1.704e-08	2.909e-06	KCNF1:67 KCNV2:145 KCNGA:148 KCNA10:185 KCNQ3:193 KCNQ2:381
KAUFFMANN_DNA_REPAIR_GENES	-0.11142718	214	2.047e-08	3.406e-06	FANCM:10 NABP2:25 BRCA1:34 MBD4:44 BRIP1:47 BRCA2:49
REACTOME_DEVELOPMENTAL_BIOLOGY	0.05429269	936	2.403e-08	3.897e-06	DSG3:46 PTK2:79 KRT80:103 COL3A1:106 COL6A1:124 KLK13:127

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Mitochondrial Diseases	-0.10383426	347	3.865e-11	3.791e-07	LRPPRC:11 TMEM126A:53 NDUFB8:57 NDUFS3:70 NDUFAF1:162 METTL7:164
Schizophrenia	0.04155697	1620	6.052e-08	2.969e-04	TRANK1:7 MROH2A:11 SHANK2:13 RGS9:16 ASAP1:17 SSTR4:21
Bipolar Disorder	0.05169587	722	2.923e-06	9.608e-03	TRANK1:7 PER2:10 SSTR4:21 SORL1:31 GABRB3:72 BRD1:83
Increased CSF lactate	-0.07865309	53	6.924e-06	1.631e-02	LRPPRC:11 NDUFS3:70 NDUFAF1:162 TMEM126B:191 NDUFA11:226 NDUFA12:451
Nuclear cataract	0.22663366	32	9.190e-06	1.631e-02	GJA3:257 CRYBA2:419 GJA8:421 CRYBB1:694 MIP:713 EPHA2:966
Primary microcephaly	-0.12496868	105	9.978e-06	1.631e-02	FANCM:10 BRIP1:47 BRCA2:49 TRMT10A:133 ORC1:137 NIN:145
Lactic acidemia	-0.11337948	121	1.709e-05	2.396e-02	LRPPRC:11 NDUFS3:70 NDUFAF1:162 TMEM126B:191 NDUFA11:226 NDUFA12:451
NADH:Q(1) Oxidoreductase deficiency	-0.24581855	25	2.105e-05	2.582e-02	NDUFS3:70 NDUFAF1:162 TMEM126B:191 NDUFA11:226 NDUFB10:598 NDUFAF2:840
Chromosome Breakage	-0.19238042	40	2.575e-05	2.807e-02	FANCM:10 BRCA1:34 BRIP1:47 BRCA2:49 DNMT3B:593 FANCE:657
Acidosis, Lactic	-0.09532604	151	5.489e-05	3.384e-02	LRPPRC:11 NDUFS3:70 NDUFAF1:162 TMEM126B:191 NDUFA11:226 ALDH7A1:314
Acute necrotizing encephalopathy	-0.27165762	19	4.154e-05	3.384e-02	NDUFS3:70 NDUFAF1:162 TMEM126B:191 NDUFA11:226 NDUFAF2:840 TIMMDC1:1627
Hyperinsulinism	0.06548972	336	4.066e-05	3.384e-02	PDE6B:26 LPIN1:65 EHM1:169 HERC2:73 OFD1:87 IGFBP3:102
Nicotinamide adenine dinucleotide coenzyme	-0.18669135	39	5.519e-05	3.384e-02	NDUFS3:70 NDUFAF1:162 NDUFA11:226 NDUFA12:451 HSD17B6:663 NDUFB6:702
Nuclear non-senile cataract	0.21673171	29	5.382e-05	3.384e-02	GJA3:257 CRYBA2:419 GJA8:421 CRYBB1:694 MIP:713 EPHA2:966
Sloping forehead	-0.13363540	79	4.100e-05	3.384e-02	CEP290:163 SETBP1:247 KAT6B:344 ASPM:352 CENPE:420 RRGRI1:480
Sudden Cardiac Death	0.16922760	48	5.041e-05	3.384e-02	EMD:132 NOS1AP:378 DSG2:402 CACNB2:477 KCNH2:731 DSC2:923
Congenital cataract	0.12269227	89	6.454e-05	3.724e-02	CRYGA:42 BFSF2:80 GJA3:257 CRYBA2:419 GJA8:421 CRYBA4:509
Leigh Disease	-0.12889514	80	6.863e-05	3.740e-02	LRPPRC:11 NDUFS3:70 NDUFA12:451 PNPT1:477 TSFM:549 MSH6:637
Autism Spectrum Disorders	0.05166186	483	1.141e-04	5.891e-02	SHANK2:13 SSTR4:21 EHM1:169 GABRB3:72 HERC2:73 SFSWAP:100
Mouth Neoplasms	0.10758715	107	1.236e-04	6.006e-02	PTK2:79 CDKN2B:288 UPP1:472 TP73:501 PTHRI:556 HMGC5:622
Channelopathies	0.16043183	47	1.430e-04	6.680e-02	SHANK3:248 KCNJ3:535 KCNH2:731 ATXN1:748 SCN1A:885 CACNA1D:1211
Psychoses, Substance-Induced	0.27159630	16	1.693e-04	7.222e-02	RGS9:16 ESRI:476 SOD2:1104 CAMK2B:1352 AKT1:1433 THRI1:1637
Severe myopia	0.11430471	91	1.674e-04	7.222e-02	PDE6B:26 IGFBP3:102 TRPM1:206 LRAP1:234 ATP6V0A2:45 ESRI:1637
CATARACT, AUTOSOMAL DOMINANT	0.33911680	10	2.047e-04	8.365e-02	BFSF2:80 GJA8:421 CRYBB1:694 MIP:713 CRYAA2:1375 CRYBB2:2332
Cataract, Pulverulent	0.33629440	10	2.311e-04	8.714e-02	BFSF2:80 GJA3:257 GJA8:421 CRYBB1:694 HSFA:1019 CRYBG:1588
Hemoglobin low	-0.08651432	152	2.398e-04	8.714e-02	FANCM:10 PARN:305 ECR6L2:347 SAR32:357 TFR2:385 ACD:426
Plaundler-Hurler Syndrome	-0.22282836	23	2.248e-04	8.714e-02	MAN2B1:156 ARSB:194 VCAN:363 GUSB:625 TBX1:668 GNPTAB:785
CATARACT, COPPOCK-LIKE	0.33085412	10	2.915e-04	9.634e-02	BFSF2:80 GJA3:257 GJA8:421 CRYBB1:694 CRYGC:1588 ISO2:1884
Diabetes Mellitus, Experimental	0.10415104	101	3.044e-04	9.634e-02	SOD1:156 MPP2:204 PKD4:238 NOS3:263 BAX:267 TIMP2:619
Allxan Diabetes	0.10415104	101	3.044e-04	9.634e-02	SOD1:156 MPP2:204 PKD4:238 NOS3:263 BAX:267 TIMP2:619
Streptozotocin Diabetes	0.10415104	101	3.044e-04	9.634e-02	SOD1:156 MPP2:204 PKD4:238 NOS3:263 BAX:267 TIMP2:619
Maternal oligohydramnios	-0.14719264	50	3.203e-04	9.818e-02	BUB1B:65 CEP290:163 TCTN3:209 PKHD1:222 INVS:248 RRGRI1:480
Night blindness, congenital stationary	0.20726891	25	3.353e-04	9.908e-02	PDE6B:26 USP11:82 TRPM1:206 TRAPPC3:395 GRM6:802 GPR34:815
Psychoses, Drug	0.22983783	20	3.741e-04	9.908e-02	RGS9:16 ESRI:476 SOD2:1104 CAMK2B:1352 AKT1:1433 THRI1:1637
Heart failure	0.04063330	683	3.441e-04	9.908e-02	BRD1:83 PRDM16:101 ADRB1:112 ITFMI:128 ARC:149 SOD1:156
Increased serum lactate	-0.10594586	95	3.655e-04	9.908e-02	LRPPRC:11 NDUFS3:70 NDUFA12:451 TSFM:549 TRMT10C:616 POLG:629
Mitochondrial Respiratory Chain Deficien	-0.17397979	35	3.703e-04	9.908e-02	PPRC1:101 PNPT1:477 HSPA14:510 HSD17B6:663 FBXL4:1096 AARS2:1099
Neutrophil count decreased	-0.12521526	67	3.985e-04	9.908e-02	ATRX:323 CSF3R:355 TRF2:385 VSP13B:508 FANCE:657 IRAK4:774
Progressive macrocephaly	-0.21312939	23	4.040e-04	9.908e-02	NDUFS3:70 NDUFAF1:162 TMEM126B:191 NDUFA11:226 NDUFAF2:840 GFAP:1464
Sinus Node Dysfunction (disorder)	0.25573923	16	3.982e-04	9.908e-02	EMD:132 TDGF1:893 CACNA1D:1211 KCN5:1286 KCNQ1:1515 AKT2:2062